

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281819

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SHIGETA; Yasushi

SECURITY ELECTRONIC SHOE FOR CASINO

Abstract

A security electronic shoe for a casino configured to house a plurality of playing cards to be drawn one by one for a game, comprises: a housing portion configured to house a stacked plurality of playing cards; an opening portion configured to expose a leading playing card among the plurality of playing cards housed in the housing portion; a draw guide portion configured to guide draw of the leading playing card from the opening portion; and a weight configured to push a last playing card of the plurality of playing cards toward the opening portion, and a bottom portion of the housing portion slopes so as to descend toward the opening portion, the weight is configured to be movable on the bottom portion of the housing portion toward the opening portion in a state pushing the last playing card toward the opening portion, according to that the last playing card moves toward the opening portion, accompanying that the leading playing card is drawn in sequence, and the housing portion has a structure in which an upper surface and at least one side surface are opened, and the plurality of playing cards can be loaded into the housing portion through the opened portion.

Inventors: SHIGETA; Yasushi (Shiga, JP)

Applicant: ANGEL GROUP CO., LTD. (Shiga, JP)

Family ID: 87136174

Appl. No.: 19/214520

Filed: May 21, 2025

Foreign Application Priority Data

JP	2022-110486	Jul. 08, 2022
JP	2022-132265	Aug. 23, 2022
JP	2023-086884	May. 26, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: A63F1/14 (20060101); A63F1/00 (20060101)

U.S. Cl.:

CPC A63F1/14 (20130101); A63F2001/001 (20130101); A63F2250/58 (20130101)

Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/218,801 filed Jul. 6, 2023, which claims the benefit of JP Pat. App. 2022-110486 filed Jul. 8, 2022, JP Pat. App. No. 2022-132265 filed Aug. 23, 2022, and JP Pat. App. No. 2023-086884 filed May 26, 2023. The contents of each application are incorporated herein by reference.

FIELD

[0002] The present invention relates to a security electronic shoe for a casino configured to house a plurality of playing cards to be drawn one by one for a game.

BACKGROUND

[0003] In a casino, a game using playing cards is performed at a gaming table. The playing cards used in the game is housed in a shoe at the gaming table. Eight decks of the playing cards are housed in the shoe, for example. A dealer draws the playing cards one by one from the shoe and progresses the game.

[0004] When the playing cards housed in the shoe are used up according to the progress of the game, the dealer loads new playing cards into the shoe. Specifically, when loading the playing cards into the shoe, the dealer inserts a cut card in a position slightly forward the last of the loaded playing cards. When the game is progressed and the cut card is drawn, the dealer stops the game after that while leaving remaining playing cards in the shoe, and removes the remaining playing cards from the shoe and loads new playing cards.

[0005] The playing cards to be loaded into the shoe are pre-shuffled and packaged. When loading the new playing cards into the shoe, it is necessary to open the package, take out several (for example, eight decks of) shuffled playing cards, and houses them in the shoe.

SUMMARY

[0006] Since the playing cards housed in the shoe are used for the game in the casino, a contents and order of the playing cards housed in the shoe must not be known to any person. That is, high security is required for the casino shoe. In particular, when opening the package, taking out the plurality of playing cards and loading them into the shoe, it is necessary to prevent a player and the dealer from knowing the contents of the playing cards.

[0007] Moreover, the game must be interrupted while the new playing cards are loaded into the shoe. In a case where this interruption is long, game participants become bored and casino sales decreases. Therefore, it is desirable that the loading of the playing cards into the shoe can be performed smoothly.

[0008] The purpose of the present invention is to provide a security electronic shoe for a casino which can load a playing card smoothly while ensuring security.

[0009] A security electronic shoe for a casino according to an aspect of the present invention is a security electronic shoe for a casino configured to house a plurality of playing cards to be drawn

one by one for a game, comprising: a housing portion configured to house a stacked plurality of playing cards; an opening portion configured to expose a leading playing card among the plurality of playing cards housed in the housing portion; a draw guide portion configured to guide draw of the leading playing card from the opening portion; and a weight configured to push a last playing card of the plurality of playing cards toward the opening portion, wherein a bottom portion of the housing portion slopes so as to descend toward the opening portion, the weight is configured to be movable on the bottom portion of the housing portion toward the opening portion in a state pushing the last playing card toward the opening portion, according to that the last playing card moves toward the opening portion, accompanying that the leading playing card is drawn in sequence, and the housing portion has a structure in which an upper surface and at least one side surface are opened, and the plurality of playing cards can be loaded into the housing portion through the opened portion.

[0010] In the above security electronic shoe for the casino, the housing portion may include a stopper at a position of the plurality of playing cards corresponding to a lower portion of the opened side surface, the stopper configured to restrict a lateral movement of the plurality of playing cards.

[0011] In the above security electronic shoe for the casino, the plurality of playing cards may be housed in the housing portion such that back surfaces thereof face the opening portion, and the weight may be configured to contact with a front surface of the last playing card such that the front surface of the last playing card cannot be seen.

[0012] The above security electronic shoe for the casino may further comprise a side cover configured to be movable between a cover position where the side cover covers the opened side surface of the housing portion and an open position where the side cover opens the opened side surface of the housing portion.

[0013] In the above security electronic shoe for the casino, the side cover may be capable of performing a rotating movement between the cover position and the open position.

[0014] In the above security electronic shoe for the casino, the side cover may be capable of performing a parallel movement between the cover position and the open position.

[0015] The above security electronic shoe for the casino may further comprise a side cover configured to be movable between a cover position where the side cover covers the opened side surface of the housing portion and an open position where the side cover opens the opened side surface of the housing portion, and a surface of the side cover opposed to the side surface may be positioned more outside than a surface of the stopper opposed to the side surface when the side cover is in the cover position.

[0016] In the above security electronic shoe for the casino, the side surface may be a side surface on a right side facing in a direction of the draw.

[0017] In the above security electronic shoe for the casino, the draw guide portion may be configured to be removable.

[0018] In the above security electronic shoe for the casino, the draw guide portion may include a metal part configured to restrain the playing card to be drawn to prevent two playing cards from being drawn at a same time.

[0019] In the above security electronic shoe for the casino, the housing portion may include an upper surface cover configured to be movable between a cover position where the upper surface cover covers upper surfaces of the plurality of playing cards and an open position where the upper surface cover opens the upper surfaces of the plurality of playing cards.

[0020] In the above security electronic shoe for the casino, the housing portion may include a holding means configured to hold the weight in a rear position of the housing portion in order to load the plurality of playing cards into the housing portion.

[0021] The above security electronic shoe for the casino may further comprise an upper surface cover configured to be fixable in a cover position where the upper surface cover covers the upper

surface of the housing portion, and the holding means may be configured to release holding of the weight according to that the upper surface cover comes to the cover position.

[0022] The above security electronic shoe for the casino may further comprise a side surface cover configured to be movable between a cover position where the side surface cover covers the opened side surface of the housing portion and an open position where the side surface cover opens the opened side surface of the housing portion, and the upper surface cover may be configured to be fixable in the cover position of the upper surface cover by positioning the side cover in the cover position of the side cover.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. 1 is a perspective view showing an electronic shoe according to a first embodiment of the present invention.

[0024] FIG. 2 is a side view showing the electronic shoe according to the first embodiment of the present invention.

[0025] FIG. 3 is a plan view showing the electronic shoe according to the first embodiment of the present invention.

[0026] FIG. 4 is a front view showing the electronic shoe according to the first embodiment of the present invention.

[0027] FIG. 5 is a view showing a state in which a plurality of playing cards is loaded into the electronic shoe according to the first embodiment of the present invention.

[0028] FIG. 6 is a view showing a structure of a weight according to the first embodiment of the present invention.

[0029] FIG. 7 is a perspective view showing the electronic shoe according to the first embodiment of the present invention.

[0030] FIG. 8 is a side view showing the electronic shoe according to the first embodiment of the present invention.

[0031] FIG. 9 is a plan view showing the electronic shoe according to the first embodiment of the present invention.

[0032] FIG. 10 is an A-A cross-sectional view of FIG. 9.

[0033] FIG. 11 is a view showing the detail of a draw guide portion according to the first embodiment of the present invention.

[0034] FIG. 12 is a view showing an example of a playing card according to the first embodiment of the present invention.

[0035] FIG. 13 is a view showing an action of a restricting bar according to the first embodiment of the present invention.

[0036] FIG. 14 is a perspective view showing an electronic shoe according to a second embodiment of the present invention.

[0037] FIG. 15 is a plan view showing the electronic shoe according to the second embodiment of the present invention.

[0038] FIG. 16 is a perspective view showing the electronic shoe according to the second embodiment of the present invention.

[0039] FIG. 17 is a perspective view showing an upper surface cover according to the second embodiment of the present invention.

[0040] FIG. 18 is a perspective view showing a relationship between the upper surface cover and a weight according to the second embodiment of the present invention.

[0041] FIG. 19 is a perspective view showing the electronic shoe according to the second embodiment of the present invention.

[0042] FIG. **20A** is a view explaining an action of a push-down protrusion according to the second embodiment of the present invention.

[0043] FIG. **20B** is a view explaining the action of the push-down protrusion according to the second embodiment of the present invention.

[0044] FIG. **21** is a perspective view showing the electronic shoe according to the second embodiment of the present invention.

[0045] FIG. **22** is a perspective view enlarged showing the front portion of the electronic shoe according to the second embodiment of the present invention.

[0046] FIG. **23A** is a view explaining an action of a backflow stopper according to the second embodiment of the present invention.

[0047] FIG. **23B** is a view explaining the action of the backflow stopper according to the second embodiment of the present invention.

[0048] FIG. **24A** is a view explaining a backflow in a prior electronic shoe.

[0049] FIG. **24B** is a view explaining the backflow in the prior electronic shoe.

[0050] FIG. **25** is a perspective view showing an electronic shoe according to a third embodiment of the present invention.

[0051] FIG. **26** is a plan view showing the electronic shoe according to the third embodiment of the present invention.

[0052] FIG. **27** is a right side view showing the electronic shoe according to the third embodiment of the present invention.

[0053] FIG. **28** is a left side view showing the electronic shoe according to the third embodiment of the present invention.

[0054] FIG. **29** is a front view showing the electronic shoe according to the third embodiment of the present invention.

[0055] FIG. **30** is a bottom view showing the electronic shoe according to the third embodiment of the present invention.

[0056] FIG. **31** is a rear view showing the electronic shoe according to the third embodiment of the present invention.

[0057] FIG. **32** is a perspective view showing the electronic shoe in a state in which an upper surface cover is removed according to the third embodiment of the present invention.

[0058] FIG. **33** is a view showing an indicating example of a result indicating display according to the third embodiment of the present invention.

[0059] FIG. **34** is a view showing the detail of a draw guide portion according to the third embodiment of the present invention.

[0060] FIG. **35** is a perspective view showing a restricting portion according to the third embodiment of the present invention.

[0061] FIG. **36** is a perspective view showing the upper surface cover according to the third embodiment of the present invention.

[0062] FIG. **37** is a perspective view showing the electronic shoe according to the third embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0063] An electronic shoe according to embodiments of the present invention is described below. In the present embodiments, the electronic shoe configured to house a plurality of playing cards to be drawn one by one by a dealer for a game at a gaming table of a casino (hereinafter simply referred to as “electronic shoe”) is described. This electronic shoe is used at the gaming table in the casino. The game may be, for example, baccarat, blackjack, and so on, but, in the present embodiments, the electronic shoe used for baccarat is described.

[0064] FIG. **1** is a perspective view showing an electronic shoe according to a first embodiment, FIG. **2** is a side view showing the electronic shoe according to the first embodiment, FIG. **3** is a plan view showing the electronic shoe according to the first embodiment, and FIG. **4** is a front view

showing the electronic shoe according to the first embodiment. The electronic shoe **100** has a rectangular shape as a whole. The electronic shoe **100** houses a plurality of playing cards such that the short side thereof are parallel to the long sides of the playing cards. The playing cards are removed to the outside of the electronic shoe **100** one by one by a draw operation by a dealer from the front end of the electronic shoe **100**.

[0065] The electronic shoe **100** includes a housing portion **10**, an opening portion **20**, a draw guide portion **30**, a weight **40**, and a main body **50**. The lower half part and rear part of the electronic shoe **100** form the main body **50**. The housing portion **10** is formed so as to extend forward contiguously from the rear end part of the main body **50**. The housing portion **10** houses a stacked plurality of playing cards. As described above, the stacked plurality of playing cards is housed in the housing portion **10** in the lateral orientation. The long sides of the playing cards are supported on the bottom surface **11** of the housing portion **10**. The playing cards are housed in the housing portion **10** such that the back surfaces of the playing cards face forward.

[0066] The bottom surface **11** of the housing portion **10** slopes so as to descend toward the opening portion **20**. The housing portion **10** has a structure in which the upper surface and left side surface (that is, the right side facing in the direction in which the playing cards are drawn) are opened. That is, the right side surface of the housing portion **10** includes a side wall **12** whose height corresponds to the height of the housed playing cards (the length of the short sides of the playing cards). At the left side surface, a side wall **14** whose height is low is also provided to stand from the bottom surface **11**, and the side wall **14** has the height corresponding to the height of the housed playing cards, at the part close to the opening portion **20**.

[0067] The part whose height is low of the side wall **14** forms a side surface opened portion of the left side surface, and an upper surface opened portion is formed to be contiguous to the side surface opened portion and opens the housing portion **10** upward. The plurality of playing cards can be loaded into the housing portion **10** through the side surface opened portion and the upper surface opened portion.

[0068] FIG. **5** is a view showing a state in which the plurality of playing cards is loaded into the electronic shoe according to the first embodiment of the present invention. The plurality of playing cards is pre-shuffled and housed in a package **P**. The package **P** is sealed, and is opened by the dealer when loaded into the electronic shoe **100**. In the package **P** shown in FIG. **5**, the upper half thereof at the plurality of playing cards is removed, and only the lower half thereof houses the plurality of playing cards.

[0069] As shown in FIG. **5**, the dealer inserts the plurality of playing cards which only half is housed in the package **P** into the housing portion **10** from the side surface opened portion and the upper surface opened portion. At this time, the dealer once places and fixes the package **P** housing the playing cards on the side surface opened portion, and then pulls out the package **P** while pushing the multiple playing cards against the side wall **12** on the back side in the housing portion **10** and align them to house them in the housing portion **10**.

[0070] The side wall **14** whose height is low is formed at a position of the plurality of playing cards corresponding to the lower portion of the side surface opened portion, and functions as a stopper to restrict a lateral movement of the housed plurality of playing cards. Thereby, even when the side surface cover **13** is in an opening state, the playing cards are neatly housed in the housing portion **10**, and so the playing cards do not move left or right, or up and down, and the playing cards do not come out from the housing portion **10**. Furthermore, as is described below, by pushing the last playing card toward the opening portion **20** by the weight **40**, the playing cards move in sequence in a forward and rearward direction without coming out of the housing portion **10**, accompanying that they are drawn out in sequence from the leading end.

[0071] The housing portion **10** is provided with the side surface cover **13** including an axis on the outside of the side wall **14** formed to be low. The side surface cover **13** rotates around this axis to perform a rotating movement between a cover position and an open position. In the cover position,

the side surface cover **13** is in an attitude to be upright, and closes the side surface opened portion to cover the housed playing cards. In the open position, the side surface cover **13** is in an attitude to be rotated further from the horizontal attitude to open the side surface opened portion.

[0072] The opening portion **20** exposes to the outside the leading playing card among the plurality of playing cards housed in the housing portion **10**. The opening portion **20** is shaped to be an arch-shape and exposes the center portion of a back surface of the playing card. The dealer can draw out the leading playing card onto the gaming table with its back surface facing up by sliding it while pressing downward on its exposed portion.

[0073] A blind **21** is formed in the opening portion **20** so as to cover over the back surface of the leading playing card exposed therethrough. The blind **21** is made of a large number of flexible thin line-shaped members, and divided into the left and right at the center. The line-shaped members are provided so as to slope in a drawing direction from the left and right to the center and from the base to the tip. The dealer can apply force to the playing card around a break of the left and right of the line-shaped members to draw out it.

[0074] The draw guide portion **30** guides a draw of the leading playing card from the opening portion **20**. Specifically, the draw guide portion **30** includes a slop portion **32** sloping so as to be continuous from the opening portion **20** to the upper surface of the gaming table, and a restricting portion **31** configured to restrict upward and sideward movements of the playing card to be drawn out and so guide it in the draw direction. The playing card is drawn out while being guided by the draw guide portion **30**, and so is drawn out onto the gaming table while the front side printed with a rank and suit faces down. Fine embossing is applied to the slop portion **32**, and thereby the front side of the playing cards is easier to slide.

[0075] FIG. **6** is a view showing a structure of the weight according to the first embodiment of the present invention. The weight **40** is used for pushing the last playing card among the plurality of playing cards toward the opening portion **20**. The weight **40** includes a roller **41** including a lateral axis. The lower end portion of the roller **41** protrudes slightly from the bottom of the weight **40**. Thereby, the weight **40** is movable in the forward and rearward direction by rotating the roller **41**.

[0076] The lateral width of the weight **40** corresponds to the lateral width of the housing portion **10**. The weight **40** includes a slope surface **42** in the front thereof. As described above, the playing cards are housed in the housing portion **10** such that the back surfaces face the opening portion **20**, and so the front surface of the last playing card contacts with the slope surface **42** of the weight **40**.

[0077] The weight **40** is movable forward and rearward on the bottom surface **11** of the housing portion **10**. As described above, the housing portion **10** slopes toward the opening portion **20**, and so the weight **40** moves to the opening portion **20** by its own weight. When the plurality of playing cards is loaded in the housing portion **10**, the slope surface **42** of the weight **40** pushes the last playing card toward the opening portion **20**. Thus, due to the action of the weight **40**, the leading playing card is always pushed from the rear side toward the opening portion **20** and exposed through the opening portion **20**.

[0078] The weight **40** can be fixed in the rear position of the housing portion **10** by magnets or an engaging mechanism. As shown in FIG. **5**, when loading the playing cards into the housing portion **10**, the weight **40** is fixed in this fixed position to form a space in which a new plurality of playing cards can be housed in the housing portion **10**. Furthermore, it may be that a sensor detects that the plurality of playing cards is housed in the housing portion **10**, and the fixation of the weight **40** by the magnets or the engaging mechanism is automatically released based on this detection of the sensor. When the fixation is released, the weight **40** moves toward the opening portion **20** on the bottom surface **11** of the housing portion **10** by its own weight, contacts the last playing card and pushes the loaded plurality of playing cards toward the opening portion **20** from the last thereof. It may be to detect that the playing cards are housed in the housing portion **10** by detecting that the side surface cover **13** closes the side surface opened portion, or that the upper surface cover **15** described below closes the upper surface opened portion, for example.

[0079] The roller **41** of the weight **40** may be a structure in which it is rotatable only in a direction to move the weight **40** forward (to the side including the slope surface **42**) and not rotatable in a direction to move the weight **40** rearward. Thereby, the weight **40** does not move rearward even if a force is applied to the weight **40** in the rearward direction when the playing cards are drawn out from the lead.

[0080] A plurality of result indicating lamps **52** is provided on the upper surface portion of the main body **50**. A result indicating button **53**, result indicating lamps **54**, and a result indicating display **55** are provided on the left side surface of the main body **50**.

[0081] The side surface opened portion and the side surface cover **13**, and the result indicating button **53** are all provided on the side facing the dealer when the electronic shoe **100** is placed on the gaming table. That is, the electronic shoe **100** is placed on the left side of the dealer, and so the side surface opened portion and the result indicating button **53** are provided on the left side facing the dealer. Thereby, it become easy for the dealer to operate the result indicating button **53** of the electronic shoe **100** placed on the gaming table and to load the new playing cards into the housing portion **10** through the side surface opened portion.

[0082] The result indicating display **55** is also provided on the left side facing the dealer. Thereby, it is possible to show a game result to the dealer without being seen by a player.

[0083] FIG. **7** is a perspective view showing the electronic shoe according to the first embodiment of the present invention, FIG. **8** is a side view showing the electronic shoe according to the first embodiment of the present invention, and FIG. **9** is a plan view showing the electronic shoe according to the first embodiment of the present invention. In the electronic shoe shown in FIGS. **7** to **9**, the side surface cover **13** closes the side surface opened portion, and the upper surface cover **15** is installed at the housing portion **10** to close the upper surface opened portion.

[0084] In this way, the side surface cover **13** covers the side surface and the upper surface cover **15** covers the upper surface, and so the playing cards housed in the housing portion **10** cannot be seen completely and security is ensured. As is described above, the side surface cover **13** is pivotable between the open position shown in FIG. **1** and the cover position shown in FIG. **7** by rotating around the rotation axis provided in the lower portion. When rotating the side surface cover **13** to be in the cover position, both the right side wall **12** and the side surface cover **13** stand upright and the upper ends thereof become a slide rail of the upper surface cover **15**.

[0085] In this way, when positioning the side surface cover **13** in the cover position, the upper ends thereof become the slide rail of the upper surface cover **15**, and so it is necessary to first stand the side surface cover **13** upright to position it in the cover position where it covers the side surface in order to position the upper surface cover **15** in the cover position, in other words, it forms a structure in which it is impossible to close the upper surface cover **15** without first closing the side surface cover **13**.

[0086] The upper surface cover **15** slides from the front to the rear on the slide rail to move to the cover position shown in FIG. **7**. Whereby, the upper surface cover **15** is installed, by which the side surface cover **13** is also fixed in the cover position.

[0087] The electronic shoe **100** may further be provided with sensors (not shown) configured to detect that the side surface cover **13** and the upper surface cover **15** are respectively in the cover position. As those sensors, magnetics or optical sensors may be used. A control device of the electronic shoe **100** outputs an error signal in a case where the sensors do not detect that the side surface cover **13** and upper surface cover **15** are in the cover positions.

[0088] The electronic shoe **100** activates a draw restricting pin **38** (described below) to prohibit the draw of the playing cards in the case where the sensors detect that the side surface cover **13** and upper surface cover **15** are not in the cover position.

[0089] FIG. **10** is an A-A cross-sectional view of FIG. **9**. As shown in FIG. **10**, the bottom portion of the housing portion **10** supports the long sides of the playing cards C, and the side wall **14** functions as the stopper and contacts the short sides of the playing cards C to restrict the lateral

movements of the playing cards C. The side surface cover **13** is contiguous to and positioned above the side wall **14** formed to be low, but the inner surface of the side surface cover **13** in the cover position is slightly outside of the inner surface of the side wall **14**. That is, a gap g is formed between the inner surface of the side wall **14** and the inner surface of the side surface cover **13** in the cover position.

[0090] By this structure, it is possible for the plurality of playing cards moving toward the opening portion **20** within the housing portion **10** by being drawn in sequence from the leading playing card to move toward the opening portion **20** without their movements being restricted by the side surface cover **13** protruding from the side wall **14**.

[0091] FIG. **11** is a view showing the detail of the draw guide portion according to the first embodiment of the present invention. In FIG. **11**, the internal structure of the restricting portion **31** is shown in a state where a portion of the restricting portion **31** is removed. The inside of the restricting portion **31** is provided with code reading sensors **33** and **34** for reading a code representing at least a number (rank) of the shuffled playing card, and card detecting sensors **35** and **36** for detecting that the playing card C is drawn out.

[0092] The restricting portion **31** of the draw guide portion **30** may be removable, and the draw guide portion **30** may also be removable from the opening portion **20**. Whereby, various types of maintenance become possible.

[0093] FIG. **12** is a view showing an example of the playing card according to the first embodiment of the present invention. In the playing card, marks M1 and M2 indicating code information are printed at positions corresponding to the code reading sensors **33** and **34**. Both marks M1 and M2 are printed with UV ink which is invisible under natural light.

[0094] Each of the code reading sensors **33** and **34** includes a lamp configured to irradiate UV light and a sensor configured to read the mark. The marks M1 and M2 representing the code information are formed in two steps at the end on the side of the short side of the card. As the code information, the two rows of marks M1 and M2 are printed over a plurality of columns. The code information is expressed by combination of presence and absence of the marks M1 and M2.

[0095] The code reading sensor **33** reads the mark M1 and the code reading sensor **34** reads the mark M2. The electronic shoe **100** reads the code information based on the combination of the read marks M1 and M2. The code reading sensors **33** and **34** obtain the code information by reading in sequence in the draw-out direction the marks M1 and M2 of the playing card being drawn out. The code information represents the rank of the playing card C. Therefore, the electronic shoe **100** can determine the rank of the playing card C being drawn out.

[0096] The electronic shoe **100** determines a game result based on the determined ranks of the cards. The game result may be a player win, banker win, or tie. When the game result indicating button **53** is pressed, the corresponding lamp of the result indicating lamps **52** is turned on according to the determined game result.

[0097] The six result indicating lamps **52** correspond to the respective game results of the player win, banker win, tic, player pair, banker pair, and banker 6, respectively. When the game result indicating button **53** is pressed again while the result indicating lamp **52** turns on, the result indicating lamp **52** is turned off to prepare a next game.

[0098] The card detecting sensors **35** and **36** are sensors of optical fiber manner configured to detect presence or absence of the playing card C. The card detecting sensors **35** and **36** are provided separately at the front and the rear in the card draw direction. When the card is being drawn out, first the card detecting sensor **35** only turns on, then the card detecting sensors **35** and **36** turn on, then the card detecting sensor **36** only turns on, and finally the card detecting sensors **35** and **36** turn off.

[0099] In a case where the card detecting sensors **35** and **36** do not turn on or off in this order, it can be considered that the playing card C is running oppositely. In this case, the electronic shoe **100** determines that it is a card draw error and turns on the corresponding error indicating lamp **51**.

[0100] The draw guide portion **30** is further provided with a restricting bar **37** configured to prevent the two playing cards from being drawn simultaneously. The restricting bar **37** is formed as a metal part. The restricting bars **37** are provided on both sides of the draw guide portion **30**.

[0101] FIG. **13** is a view showing an action of the restricting bar according to the first embodiment of the present invention. A gap through which only one playing card can pass is formed between the restricting bar **37** and the slope portion **32**. The restricting bar **37** acts so as to press from the upper side the playing card **C** being drawn while sliding down the slope portion **32**.

[0102] In FIG. **13**, even when, accompanying that the leading playing card **C1** is drawn out, the second playing card **C2** is pulled to follow therewith and moves forward, the leading playing card **C1** passes between the restricting bar **37** and the slope portion **32** and so the leading playing card **C1** is pressed against the slope portion **32**, and a second playing card **C2** cannot move forward any further, whereby preventing two cards draw.

[0103] The draw guide portion **30** is further provided with a draw restricting pin **38**. The draw restricting pin **38** is buried under the slope portion **32** in a situation in which it is permitted to draw the playing card **C**, and protrudes from the slope portion **32** to restrict a draw of the playing card **C** in a situation in which it is prohibited to draw the playing card **C**.

[0104] In baccarat, the game result is determined by the ranks of the playing cards drawn in sequence. It is determined whether fifth and sixth cards should be drawn based on the ranks of the first to fourth cards. Therefore, the electronic shoe **100** can judge the end of the game based on the ranks of the playing cards **C** drawn in sequence.

[0105] When the game ends, the dealer presses the result indicating button **53** to turn on the result indicating lamp **52**, and further presses the result indicating button **53** to turn off the result indicating lamp **52**, and the next game starts. After the game ends (the game result is decided), it is prohibited to draw the playing card until the result indicating button **53** is pressed twice and the next game starts.

[0106] In such a situation in which the draw is prohibited, the draw restricting pin **38** protrudes to prevent the prohibited draw of the playing card. The draw restricting pin **38** may be buried within the slope portion **32** even in the situation in which the draw is prohibited, and may protrude to prevent the draw of the playing card according that the upstream card detecting sensor **35** detects the card in the situation in which the draw is prohibited. The draw restricting pin **38** further activates when the playing card is overdrawn, when the playing card is transferred backward, when the playing card is stopped in the opening portion **20**, and so on.

[0107] Whereby, it becomes unnecessary that action that the draw restricting pin **38** protrudes at each time when the game ends and retracts at each time when the game starts, it can be prevented for mechanical noises generated by that the draw restricting pin **38** moves up and down from bothering the dealer and player, and it can be reduced that possibility of malfunction caused by numerous actions.

[0108] In the above embodiment, the side surface cover **13** has the structure in which it can perform the rotating movement between the cover position and the open position, but the side surface cover **13** may have a structure in which it can perform a parallel movement between the cover position and the open position. Moreover, the side surface cover **13** may be installable in the cover position and also removable from the cover position.

[0109] An electronic shoe according to a second embodiment of the present invention is described below. In the electronic shoe according to the second embodiment, the same reference mark is given to the same structure as that of the electronic shoe according to the first embodiment, and explanations thereof are omitted appropriately. FIG. **14** is a perspective view showing the electronic shoe according to the second embodiment of the present invention. In FIG. **14**, the electronic shoe in a state in which an upper surface cover is installed is shown. The basic structure of the electronic shoe **101** according to this embodiment is the same as that of the electronic shoe **100** according to the first embodiment. The electronic shoe **101** includes a housing portion **10**, an

opening portion **20**, a draw guide portion **30**, a weight **40** and a main body **50**.

[0110] FIG. **15** is a plan view showing the electronic shoe according to the second embodiment of the present invention, and FIG. **16** is a perspective view showing the electronic shoe according to the second embodiment of the present invention. In FIGS. **15** and **16**, the electronic shoe in a state in which the upper surface cover removed is shown. A bottom surface **11** of the electronic shoe **101** is provided with weight stoppers **111** at a position of about $\frac{1}{4}$ of the total length of the housing portion **10** from the rear end of the housing portion **10**. The weight stoppers **111** are provided one by one on the right and the left. The left and right weight stoppers **111** are formed contiguously from side walls **12** and **14**, respectively. The height of the weight stopper **111** is about 1 mm.

[0111] When the weight **40** is placed at the rear end of the housing portion **10** in the state in which the upper surface cover **15** is removed, the front end of a slope surface **42** contacts the weight stopper **111**, and so it is restricted to fall down on the slope surface of the bottom surface **11** and remains on the rear end side (an upper portion holding position) of the housing portion **10**. The dealer can collect the remaining playing cards and load new playing cards in this state, that is, the state in which the weight **40** does not press the playing cards forward.

[0112] Each of the outside of the upper end of the side surface cover **13** and the outside of the upper end of the side wall **12** of the housing portion **10** are provided with engaging protrusions **16** protruding outwardly. The engaging protrusions **16** are provided above a slide rail **17**, and engaging protrusions **154** of the upper surface cover **15** described below are inserted between the slide rail **17** and these engaging protrusions **16**, and so the upper surface cover **15** is installed and fixed to the housing portion **10**.

[0113] The distance from the weight stopper **111** to the rear end of the housing portion **10** is designed so as to secure a gap **g2** to the extent that a finger can be inserted between the rear end of the upper surface of the weight **40** and the rear end of the housing portion **10** in the state in which the front end of the slope surface **42** of the weight **40** is restricted from descending the slope surface of the bottom surface **11** by the weight stopper **111**. By securing this gap **g2**, the dealer can pick up the weight **40** with the finger and position it in the upper portion holding position.

[0114] FIG. **17** is a perspective view showing the upper surface cover according to the second embodiment of the present invention. A plurality of ribs **152** are formed in the longitudinal direction on an inner surface **151** of the upper surface cover **15**. The rib **152** increases the rigidity of the upper surface cover **15**, and, in addition, performs a function for restraining the playing cards housed in the electronic shoe **101** from rising too much due to the action of the weight **40** when it is installed to the electronic shoe **101**. That is, the rib **152** contact the upper surfaces (upper long sides) of the playing cards housed in the housing portion **10** and pressed forward by the weight **40** when the upper surface cover **15** is installed to the electronic shoe **101**.

[0115] Moreover, the rear end of the inner surface **151** is provided with a push-down protrusion **153**. In the present embodiment, the push-down protrusion **153** is provided at a position closer to one side of the width direction, but the push-down protrusion **153** may be provided in the center of the width direction, a plurality of push-down protrusions **153** may be provided in the width direction, and it may be provided over the whole of the width direction.

[0116] Furthermore, longitudinal walls are formed on the long sides on both sides of the upper surface cover **15**, and the lower ends of the longitudinal walls are provided with a plurality of engaging protrusions **154** toward the inner side of the width direction. These engaging protrusions **154** are provided in positions where these engage with the engaging protrusions **16** provided in the housing portion **10** when the upper surface cover **15** is positioned in a position where it completely covers the housing portion **10** (that is, the position of the upper surface cover **15** in FIG. **14**, hereinafter referred to as "cover position").

[0117] FIG. **18** is a perspective view showing a relationship between the upper surface cover and the weight according to the second embodiment of the present invention. FIG. **18** shows a positional relationship between the weight **40** and the upper surface cover **15** when the weight **40** is

in the upper portion holding position and the upper surface cover **15** is about to be installed in the cover position. At this time, the push-down protrusion **153** contacts with the rear end portion of the upper surface **43** of the weight **40**.

[0118] The rear end portion of the upper surface **43** of the weight **40** includes an cave portion **431** formed to extend from the upper surface **43**. The dealer can grasp the weight **40** or lift the rear end side thereof by hooking the fingertip on the cave portion **431**.

[0119] FIG. **19** is a perspective view showing the electronic shoe according to the second embodiment of the present invention. FIG. **19** shows a state in which new playing cards are loaded in a state in which the weight **40** is held in the upper portion holding position. The dealer houses the playing cards in a state in which these lay such that the plurality of playing cards is loaded in the whole space in front of the weight **40** held in the upper portion holding position in the housing portion **10**. Whereby, the upper surfaces of the plurality of playing cards are positioned in relatively low positions and do not interfere with the installation of the upper surface cover **15** to the housing portion **10**.

[0120] FIGS. **20A** and **20B** are views explaining the action of the push-down protrusion according to the second embodiment of the present invention. FIGS. **20A** and **20B** show a position and a movement of the push-down protrusion **153** when the upper surface cover **15** slides from the front to the rear on the slide rail **17**. When the upper surface cover **15** is positioned at the more front than the cover position, the push-down protrusion **153** is positioned in a position apart from the upper surface **43** of the weight **40**, as shown in FIG. **20A**. When the upper surface cover **15** slides from the front to reach the cover position, the push-down protrusion **153** pushes down the rear end of the upper surface **43** of the weight **40**.

[0121] By the action of the push-down protrusion **153**, the front end of the weight **40** is lifted up using the rear end of the bottom surface as a fulcrum (X mark in FIG. **20B**) by the principle of leverage. Whereby, the front edge of the weight **40** crosses over the weight stopper **111**. In this state, a movement of the weight **40** is not restricted by the weight stopper **111**, and so the weight **40** slides to descend on the sloping bottom surface **11** by its own weight. At this time, a weight portion **44** of the weight **40** (referred to FIG. **18**) is arranged in the front of the weight **40** (below the slope surface **42**), and so the weight **40** slides downward smoothly crossing over the weight stopper **111**.

[0122] The diameter of both end portions corresponding to the weight stopper **111** in the roller **41** may be designed to be smaller corresponding to the thickness of the weight stopper **111**. Whereby, it is not necessary for the roller **41** to cross over the weight stopper **111**, the roller **41** is not stopped by the weight stopper **111**, and it becomes possible that the smooth movement on the weight stopper **111**.

[0123] In this way, by the action of sliding the upper surface cover **15** to the cover position, it is released that the state in which the weight **40** held in the upper portion holding position by the weight stopper **111**, and the weight **40** slides downward to press the last playing card toward the front. That is, the holding of the weight **40** is released according that the upper surface cover **15** comes to the cover position. Whereby, the plurality of playing cards rises up at the angle of the slope surface **42** of the weight **40**, but this rising of the playing cards is restrained by the rib **152** of the installed upper surface cover **15**.

[0124] Previously, a weight is slid forward by its own weight in a state in which an upper surface cover is opened, and, in a state in which a plurality of playing cards rises up along a slope surface of the weight thereby, the upper surface cover is installed. In this case, the upper surface cover pushes down the upper surfaces of the standing-up plurality of playing cards against the forward force caused by weight's own weight, and thereby, the upper surface cover is installed to the housing portion. In this case, it must be adapted that a manner that the upper surface cover is installed from the upper side toward the lower side with respect to the housing portion and fixed by magnets and the like, and, when the plurality of playing cards raises up by the weight's own weight, the upper surface cover may be lifted up thereby.

[0125] In contrast, in the present embodiment, it is adapted that a structure in which the upper surface cover **15** is installed to the housing portion **10** by sliding it, and the upper surface cover **15** is configured to be installed to the housing portion **10** by engaging the engaging protrusions **16** (referring to FIG. **16**) provided on the slide rail **17** of the housing portion **10** with the engaging protrusions **154** of the upper surface cover **15** by sliding it with respect to the housing portion **10**, and so the upper surface cover **15** is prevented from lifting up. Moreover, the movement restriction of the weight **40** by the weight stopper **111** is released to slide it forward by the action of sliding the upper surface cover **15** in such a manner, and so it is possible to press the last of the plurality of playing cards toward the front at the same time when the upper surface cover **15** is installed in the cover position or immediately after the upper surface cover **15** is installed in the cover position, and to certainly restrain the rise of the plurality of playing cards due to the weight's own weight **40** by the ribs **152** of the upper surface cover **15**.

[0126] FIG. **21** is a perspective view showing the electronic shoe according to the second embodiment of the present invention. FIG. **21** shows a state in which the weight **40** crosses over the weight stopper **111** and presses the plurality of playing cards forward from the last. As mentioned above, such a state is realized by installing the upper surface cover **15**, but FIG. **21** shows the state in which the weight **40** pushes the plurality of playing cards forward in a state in which the upper surface cover **15** is removed.

[0127] FIG. **22** is a perspective view enlarged showing the front portion of the electronic shoe according to the second embodiment of the present invention. FIG. **22** shows in a state in which some parts on the left side are removed to expose backflow stoppers **112**. The front end of the housing portion **10** is provided with two left and right backflow stoppers **112** so as to cover the upper surface opened portion. The backflow stopper **112** restricts that the leading playing card housed in the housing portion **10** flows back due to an operation of the dealer.

[0128] The backflow stopper **112** includes a base end portion **1121** extending rearward and upward in a direction substantially parallel to plane directions of the housed playing cards, a backflow restricting portion **1122** contiguous from the tip end of the base end portion **1121** and bent downward at a substantially right angle, and a restraining portion **1123** contiguous from the tip end of the backflow restricting portion **1122** and extending backward so as to be along the upper surface of the housed playing cards.

[0129] FIGS. **23A** and **23B** are views explaining an action of the backflow stopper according to the second embodiment of the present invention. FIG. **23A** shows a normal state. In FIG. **23A**, the bottom edge of each playing card contacts with the bottom surface **11**. As shown in FIG. **23B**, when the leading (front end) playing card is pushed up with the finger, the upper end of the leading playing card rises along the base end portion **1121** of the backflow stopper **112** and hits against the backflow restricting portion **1122**. Whereby, it is prevented that the backflow of the leading playing card.

[0130] As shown in FIG. **23B**, a slight rise (backflow) is allowed to the extent that the upper end of the leading playing card hits the backflow restricting portion **1122**, but the leading playing card cannot be moved to the extent that the second playing card can be drawn out.

[0131] FIGS. **24A** and **24B** are views explaining a backflow in a prior electronic shoe. In an example in FIGS. **24A** and **24B**, the upper surfaces of a plurality of playing cards housed in the housing portion are covered by a flat ceiling **150**. In such a case, as shown in FIG. **24A**, when a leading playing card is pushed up with the finger, the front end of the leading playing card enters along the ceiling **150** into a space between the ceiling **150** and the upper ends of a second and rear playing cards to rise, and the lower end portion of the second playing card is exposed. Whereby, the finger can contact the second playing card, and skip the leading playing card and draw the second playing card, as shown in FIG. **24B**.

[0132] In contrast, in the present embodiment, the backflow of the leading playing card is restricted by the backflow stopper **112**, and so it is impossible to contact and draw out the second playing

card, and it is prevented that the illicitness that the playing card is drawn in a different order from the original order.

[0133] In the second embodiment of the present invention, the following techniques are disclosed.

[0134] (1) A security electronic shoe for a casino configured to house a plurality of playing cards to be drawn one by one for a game, comprising: a housing portion configured to house a stacked plurality of playing cards; an opening portion configured to expose a leading playing card among the plurality of playing cards housed in the housing portion; a draw guide portion configured to guide draw of the leading playing card from the opening portion; a weight configured to push a last playing card of the plurality of playing cards toward the opening portion; and an upper surface cover configured to be slidable between a cover position where the upper surface cover covers upper surfaces of the plurality of playing cards and an open position where the upper surface cover opens the upper surfaces of the plurality of playing cards, wherein the bottom portion of the housing portion slopes so as to descend toward the opening portion, the weight is configured to be movable on the bottom portion of the housing portion toward the opening portion in a state pushing the last playing card toward the opening portion, according to that the last playing card moves toward the opening portion, accompanying that the leading playing card is drawn in sequence, the bottom portion of the housing portion includes a weight stopper configured to hold the weight in a holding position at an upper portion by contacting a front end of the weight, and the weight is configured that a holding state of the weight in the holding position is released by positioning the upper surface cover in the cover position and the weight moves toward the opening portion by its own weight to push the last playing card toward the opening portion.

[0135] (2) The security electronic shoe for the casino according to (1), wherein the upper surface cover includes a push-down protrusion on an inner surface, and when the upper surface cover is positioned in the cover position, the push-down protrusion pushes down a rear end portion of an upper surface of the weight held in the holding position by the weight stopper, the front end of the weight crosses over the weight stopper by principle of leverage, and the weight moves toward the opening portion by its own weight to push the last playing card toward the opening portion.

[0136] (3) The security electronic shoe for the casino according to (1) or (2), wherein the weight includes an cave portion extending rearward from the upper surface.

[0137] (4) The security electronic shoe for the casino according to any of (1) to (3), wherein weight stoppers are provided on both sides of the bottom portion of the housing portion.

[0138] (5) The security electronic shoe for the casino according to any of (1) to (4), wherein the weight stopper is provided at a position where a gap is secured between a rear end of the weight and a rear end of the housing portion when the weight is held in the holding position.

[0139] (6) A security electronic shoe for a casino configured to house a plurality of playing cards to be drawn one by one for a game, comprising: a housing portion configured to house a stacked plurality of playing cards; an opening portion configured to expose a leading playing card among the plurality of playing cards housed in the housing portion; a draw guide portion configured to guide draw of the leading playing card from the opening portion; and a weight configured to push a last playing card of the plurality of playing cards toward the opening portion, wherein an upper portion of a front end of the housing portion is provided with a backflow stopper configured to restrict a backflow of the leading playing card housed in the housing portion.

[0140] (7) The security electronic shoe for the casino according to (6), wherein the backflow stopper is provided so as to be covered over an opened portion of an upper surface of the housing portion.

[0141] (8) The security electronic shoe for the casino according to (6) or (7), wherein the backflow stopper includes a base end portion extending rearward and upward, and in a direction substantially parallel to a plane direction of the playing cards housed in the housing portion, and a backflow restricting portion contiguous from a tip end of the base end portion and bent downward at a substantially right angle.

[0142] An electronic shoe according to the third embodiment of the present invention is described below. FIGS. **25-31** are a perspective view, a plan view, a right side view, a left side view, a front view, a bottom view, and a rear view showing the electronic shoe according to the third embodiment of the present invention. FIG. **32** is a perspective view showing the electronic shoe in a state in which an upper surface cover is removed according to the third embodiment of the present invention.

[0143] In the electronic shoe according to the third embodiment, the same reference mark is given to the same structure as that of the electronic shoe according to the second embodiment, and explanations thereof are appropriately omitted. A basic structure of the electronic shoe **102** according to the present embodiment is the same as that of the electronic shoe **101** according to the second embodiment. The electronic shoe **102** includes a housing portion **10**, an opening portion **20**, a draw guide portion **30**, a weight **40**, and a main body **50**.

[0144] In the electronic shoe **102** according to the present embodiment, comparing with the electronic shoe **101** according to the second embodiment, in corner portions of the front end of the draw guide portion **30**, the corners are removed, and those are shaped to be rounded. Whereby, it is easier that an open-hand operation that a dealer draws a playing card C from the electronic shoe **102** and simultaneously turns over the playing card C to face the front side up. That is, when the dealer draws out the playing card C with the index finger of the right hand while contacting the ball of the thumb of the right hand with the corner portion on the side of the dealer, the dealer contacts the front side of the playing card C with the thumb and the back side of the drawn card C with the index finger, and can pick up the playing card C, and can easily perform an action of turning over the playing card C subsequently to it.

[0145] The electronic shoe **102** in the present embodiment includes magnets respectively built into the rear end of an upper surface cover **15** and the rear end portion of the housing portion **10** which contacts the rear end of the upper surface cover **15** when the upper surface cover **15** is installed. Whereby, a position of the upper surface cover **15** is held by magnetic force of the magnets in a state in which it is installed in the electronic shoe **102**. The upper surface cover **15** can be easily removed by sliding the upper surface cover **15** against the magnetic force of the magnets.

[0146] In the electronic shoe **102** according to the present embodiment, a metal reinforcing plate is built into each of a side surface cover **13** and a side wall portion integrally forming the side surface together with the side surface cover **13**. Whereby, when gripping the side surface cover **13** and the portion of a side wall **12** opposite thereto to lift the electronic shoe **102**, even in the case where inward pressure is applied to the side surface cover **13**, it is possible to reduce deformation of the side surface cover **13**.

[0147] In the electronic shoe **102** according to the present embodiment, a protrusion **157** extending in the forward and reward direction is formed on the outside of the side surface of the upper surface cover **15**. Whereby, it is facilitated to hang the finger on this protrusion **157** to lift the electronic shoe **102** in a state in which the upper surface cover **15** is installed.

[0148] As shown in FIG. **31**, the back surface of the electronic shoe **102** includes a connector **62** for connecting with an external device and a power switch **61** of the electronic shoe **102**.

[0149] As shown in FIG. **32**, chamfering is performed to outer portions **132** of the upper ends of the front and rear ends of the side surface cover **13** and portions **141** of a side wall **14** opposite to the outer portions **132** of the side surface cover **13**. Whereby, it is prevented that the inner side surface of the upper surface cover **15** is caught by a step at a boundary between the upper end of this side surface cover **13** and the side wall **14** to hinder a smooth sliding in a process in which the upper surface cover **15** slides to the closed position.

[0150] FIG. **33** is a view showing an indicating example of a result indicating display according to the third embodiment of the present invention. The result indicating display **55** according to the present embodiment indicates information in two rows. Specifically, in the result indicating display **55**, as a game result, ranks and suits of two playing cards C which is banker hand are indicated in

the upper row on the left side and a number of an units digit of a total value of ranks of two playing cards C which is banker hand is indicated in the lower row, and ranks and suits of two playing cards C which is player hand are indicated in the upper row on the right side and a number of an units digit of a total value of ranks of two playing cards C which is player hand is indicated in the lower row.

[0151] The electronic shoe **102** according to the present embodiment is provided with, as a result indicating lamp **54**, a lamp for indicating banker 6 as well as lamps for indicating the respective results of player win, banker win, and tic. Banker 6 means that a hand of banker wins by 6. In this case, even when betting on the banker win, an amount of chips to be paid out is a half amount of a bet amount. In a case where a bet area is provided with a bet area for betting on that the banker wins by 6 (called “super 6”, “lucky 6”, and so on), payment is performed for chips bet thereon at a multiplier greater than 1 time (for example, 24 times).

[0152] In this may, when the banker 6 occurs, the dealer must perform a different payment operation from that at normal time. Therefore, the electronic shoe **102** according to the present embodiment is provided with a lamp for indicating that the banker 6 occurs as one of the result indicating lamps **54**.

[0153] FIG. **34** is a view showing a detail of the draw guide portion according to the third embodiment of the present invention. In FIG. **34**, an inside structure of a restricting portion **31** is shown in a state in which the restricting portion **31** is removed. Each of the front end portions of the side walls **12** and **14** includes a restraining portion **18** for restraining lifting of housed playing cards C. The lower surface of this restraining portion **18** has a curved shape convex downward. By this structure, the front end of a leading playing card C stops in front of a restricting bar **37**, and it is not that the leading playing card C housed in the housing portion **10** is pressed against the restricting bar **37** by its own weight and the pressure of the weight **40**.

[0154] By such structure, the playing card C does not hinder when removing the restricting portion **31** as is shown in FIG. **34** and maintaining the restricting bar **37**. That is, for the height of the restricting bar **37**, a precise adjustment is necessary in order to prevent two cards draw. Therefore, the adjustment may be necessary in a process of using the electronic shoe **102**, and, even in such case, the leading playing card C does not contact with the restricting bar **37**, and so it is possible to adjust the restricting bar **37** without removing the playing cards C from the housing portion **10**.

[0155] As shown in FIG. **34**, the electronic shoe **102** according to the present invention is provided with a sensor cover **39** covering code reading sensors **33** and **34** and card detecting sensors **35** and **36**. The periphery portion of the sensor cover **39** is provided with a packing in order to prevent paper dust of the playing card C from entering the interior. The sensor cover **39** is formed of transparent material.

[0156] In order to prevent the paper dust of the playing card C from entering the code reading sensors **33** and **34** and card detecting sensors **35** and **36**, a sponge-like dustproof member or seal member surrounds the peripheries of sensor chips on a sensor substrate.

[0157] FIG. **35** is a perspective view showing the restricting portion according to the third embodiment of the present invention. The restricting portion **31** has a generally fork-shaped shape, and the lower surfaces of the front end portions thereof are provided with engaging claws **22**. These engaging claws **22** engage claw receiving holes **19** formed at the front end portions of the side walls **12** and **14**. These engaging claws **22** of the restricting portion **31** engage with the claw receiving holes **19** of the side walls **12** and **14**, and so it is prevented that the front end portions of the side walls **12** and **14** are spread to the outside of the left and right and the front end portions of the restricting portion **31** are lifted up.

[0158] FIG. **36** is a perspective view showing the upper surface cover according to the third embodiment of the present invention. A plurality of ribs **152** is formed on the inner surface **151** of the upper surface cover **15**. In addition to this, guides **155** extending in the longitudinal direction is formed at positions close to the left and right side walls **156** on the inner surface **151**. When the

upper surface cover **15** is installed to the electronic shoe **102**, the upper ends of the side walls **12**, **14** and the side surface cover **13** are inserted into gaps between the guides **155** and the left and right side walls **156**.

[0159] That is, in the state in which the upper surface cover **15** is installed, the upper ends of the side walls **12**, **14** and side surface cover **13** fit into and fixed in the gaps between the guides **155** formed on the inner surface **151** of the upper surface cover **15** and the side wall **156**. Therefore, in the state in which the upper surface cover **15** is installed, the side walls **12**, **14** and side surface cover **13** and the upper surface cover **15** are firmly fixed, and, together with the action of the metal reinforcing plate described above, it is possible to reduce deformation when the electronic shoe **102** is lifted.

[0160] In baccarat game in which the electronic shoe **102** is used, there are a type of a game operation called squeeze baccarat in which a player is permitted to open a playing card C drawn from the electronic shoe **102**, and a type of a game operation called open hand baccarat in which the dealer oneself opens a playing card C drawn from the electronic shoe **102**. In the case of the open hand baccarat, the dealer opens the playing card C drawn from the electronic shoe **102**. In the case of the open hand baccarat, the dealer draws a playing card C from the electronic shoe **102** and simultaneously opens the playing card C and places it on the gaming table.

[0161] FIG. **37** is a perspective view showing the electronic shoe according to the third embodiment of the present invention. In the case of the above open hand, the bottom surface of the electronic shoe **102** is installed with bottom raising parts **71** and **72** raising the bottom surface of the electronic shoe **102**. Whereby, the height of the upper surface of the draw guide portion **30** on which the playing card C to be drawn slides from a table surface of the gaming table become higher, and so it become easier for the dealer to pick up the drawn card.

[0162] The bottom raising part **71** is installed to the front end portion of the bottom surface, and the two bottom raising parts **72** are installed to the left and right of the rear end portion of the bottom surface. The bottom raising part **71** may be installed to the bottom surface by a screw, and the bottom raising part **72** may be installed to the bottom surface by itself as a screw.

[0163] In the electronic shoe **102** according to the present embodiment, as is in the electronic shoe **101** according to the second embodiment, when the upper surface cover **15** is closed, holding of the weight **40** at the upper portion of the housing portion **10** is released, and the weight **40** slides down in the housing portion **10** to press the last of the housed playing cards C forward. In this way, it is that a structure in which the weight **40** slides down by installing the upper surface cover **15**, and so the user cannot visually confirm whether the weight **40** become indeed in a state in which it slides down and pushes the playing cards C forward.

[0164] Therefore, an observation window may be provided on the upper surface cover **15**. This observation window may be provided at the upper portion of the upper surface cover **15** and it may be possible to observe the weight **40** held at the upper end portion of the housing portion **10**. Whereby, when the upper surface cover **15** is installed to the electronic shoe **102** and the weight **40** slides downward, the user cannot observe the weight **40** through the observation window, and so check that the weight **40** slid downward.

[0165] Instead of the observation window, it may be used that a sensor configured to detect that the weight **40** slid from a holding position. For example, it may be to provide a sensor formed of an LED light and a photodiode sensor at a position corresponding to the weight **40** and detect presence or absence of the weight **40** in the holding position.

[0166] A computer is built in the main body **50** of the electronic shoe **102** according to the present embodiment. This computer includes an arithmetic processor, memory, and storage device, and, as a processing portion of the electronic shoe **102**, activates by a program according to the present embodiment and performs the following information processing. First, the processing portion stores information on ranks of drawn cards and game results as log information in the storage device together with information on a date and time of the game. Moreover, when the processing

portion detects that the upper surface cover **15** is opened or closed as well as when some error occurs, it stores those events as the log information in the storage device together with the information of the date and time.

[0167] Furthermore, in the processing portion, analyzing software configured to analyze this log information may be ran. This analyzing software may, for example, perform statistical processing on the game results or on the ranks of the drawn playing cards C.

[0168] The processing portion is connected to an external computer through the connector **62**. Whereby, the external computer can read out various log information stored in the storage device by connecting to the processing portion through the connector **62**. Therefore, the processing portion has a security function in order to allow to export the log information only to trusted external computer. Specifically, an ID and password are set to the processing portion, and only when these ID and password are input from connected external computer, it communicates with the external computer.

[0169] The processing portion obtains information on the ranks of the drawn playing cards C from the code reading sensors **33** and **34**. As described above, the two rows of marks M1 and M2 are printed on the playing card C by UV ink, and the code reading sensors **33** and **34** read these marks M1 and M2 to obtain the information on the ranks. At this time, it is set that a threshold value by which the sensors determine presence or absence of marks in marks M1 and M2.

[0170] The processing portion runs a program for teaching the threshold value for determining the presence or absence of these marks for the code reading sensors **33** and **34**. Previously, a threshold value is set by operating each of a plurality of sensors. In the electronic shoe **102** according to the present embodiment, the teaching can be performed at once for this plurality of code reading sensors **33** and **34** by the processing portion or the external computer connected to the processing portion.

[0171] In this teaching, the playing card C is set at a position where the mark M1 is read by the code reading sensor **33**, a reading value of the code reading sensor **33** is obtained, and the threshold value of the code reading sensor **33** is set based on this. In the same way, the playing card C is set at a position where the mark M2 is read by the code reading sensor **34**, a reading value of the code reading sensor **34** is obtained, and the threshold value of the code reading sensor **34** is set based on this.

[0172] In the teaching, the processor may alternatively obtain a change in a reading value of the code reading sensors **33** and **34** (values of areas in which the marks M1 and M2 are present and absent) while drawing the playing card C, and set the threshold value based on this reading value.

[0173] The processing portion or the external computer connected to the processing portion also perform teaching for color tone of the playing card C. In this teaching, for example, about 20 playing cards C are drawn, the code reading sensors **33** and **34** perform reading during this, and so the processing portion determines the color tone of the playing card C and adjust the white balance.

Claims

1. A security electronic shoe for a casino configured to house a plurality of playing cards and to read the rank and suit of cards that are drawn one by one for a game, comprising: a housing portion configured to house a stacked plurality of playing cards; an opening portion configured to expose a leading playing card among the plurality of playing cards housed in the housing portion; a draw guide portion configured to guide draw of the leading playing card from the opening portion; a weight configured to push a last playing card of the plurality of playing cards toward the opening portion; a card drawing restricting member configured to restrict the leading playing card to be drawn from the opening portion so that no card is drawn out; and a two-card drawing restricting member configured to restrict two playing cards to be drawn from at a same time so that the cards are correctly drawn one at a time, wherein a bottom portion of the housing portion slopes so as to

descend toward the opening portion, wherein the weight is configured to be movable on the bottom portion of the housing portion toward the opening portion in a state pushing the last playing card toward the opening portion, according to that the last playing card moves toward the opening portion, accompanying that the leading playing card is drawn in sequence, wherein the draw guide portion includes a two-card drawing restricting member configured to restrain the playing card to be drawn to prevent two playing cards from being drawn at a same time, and wherein, depending on an effect of the card restriction member, drawing of the leading playing card is restricted, and depending on an effect of the two-card drawing restriction member, drawing of a following playing card is restricted.

2. The security electronic shoe for casino according to claim 1, further comprising a reading portion configured to read at least a rank of cards drawn from the opening portion, wherein the card drawing restriction member is configured to stop drawing of the leading playing card before the rank of the leading playing card is read by the reading portion, thereby preventing the rank of the leading playing card restricted by the card restriction member from being read.

3. The security electronic shoe for the casino according to claim 1, wherein the two-card drawing restricting member is a restricting bar, disposed above the bottom portion of the housing portion with a gap through which only one playing card can pass, the restricting bar being configured to press from the upper side the playing card being drawn and sliding down the bottom portion.

4. The security electronic shoe for the casino according to claim 2, wherein the two-card drawing restricting member has a rounded shape on a side that faces the leading playing card, thereby the leading playing card is led into the gap.

5. The security electronic shoe for the casino according to claim 1, wherein the restricting member is a restricting bar, disposed above the bottom portion of the housing portion so that when a leading playing card is drawn out and a following playing card moves forward being dragged by the leading playing cards, the leading playing card passes through a gap between the two-card drawing restricting bar and the bottom portion and so the leading playing card is pressed against the bottom portion by the restricting bar, and the following playing card is prevented from moving forward toward the gap, thereby preventing two cards from being drawn.

6. The security electronic shoe for the casino according to claim 1, wherein the two-card drawing restricting member is provided the both side of the draw guide portion.

7. The security electronic shoe for the casino according to claim 1, further comprising: an upper surface cover configured to be attached to the top of the housing portion and open the top of the housing portion by being removed from the housing portion; and a sensor configured to output a signal when detecting the removing of the upper surface cover from the housing portion.

8. The security electronic shoe for the casino according to claim 1, further comprising a backflow stopper configured to restrict that the leading playing card housed in the housing portion from flowing back. so that a following playing card is not exposed from the opening portion due to the backflow of the leading playing card.
